

# Gears, speed and direction

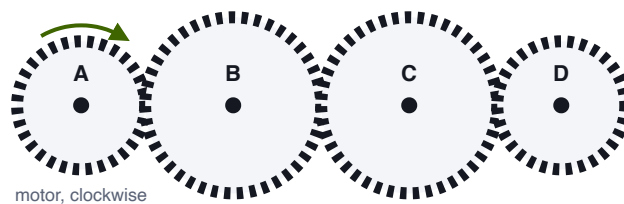
Name: \_\_\_\_\_ Class: \_\_\_\_\_ Date: \_\_\_\_\_

Meshing gears turn in **opposite directions**.  $\text{output speed} = \text{input speed} \times \text{driver teeth} \div \text{driven teeth}$

A small gear driving a big gear turns it **slower** but with more **torque** (turning force).

## Part A · Gears

**A1** Gear A is on a motor that turns clockwise.



a) Which way does each of gears B, C and D turn? [2]

b) How could you make gear D turn clockwise? [1]

**A2** A 24-tooth gear on a motor turning at 20 rpm drives an 8-tooth gear.

a) How many times faster does the 8-tooth gear turn? [1]

b) What is its speed in rpm? [1]

**A3** An 8-tooth gear on a motor turning at 30 rpm drives a 32-tooth gear on a winch.

a) What is the winch's speed? [1]

b) The motor gives 2 N·m of torque. Ignoring friction, what torque does the winch get? [2]

**A4** In Gear Garage, two gears mesh when their axles are  $(\text{teeth of one} + \text{teeth of the other}) \div 8$  holes apart. How far apart must these be?

a) two 16-tooth gears   b) an 8-tooth and a 24-tooth gear   c) an 8-tooth and a 32-tooth gear [3]

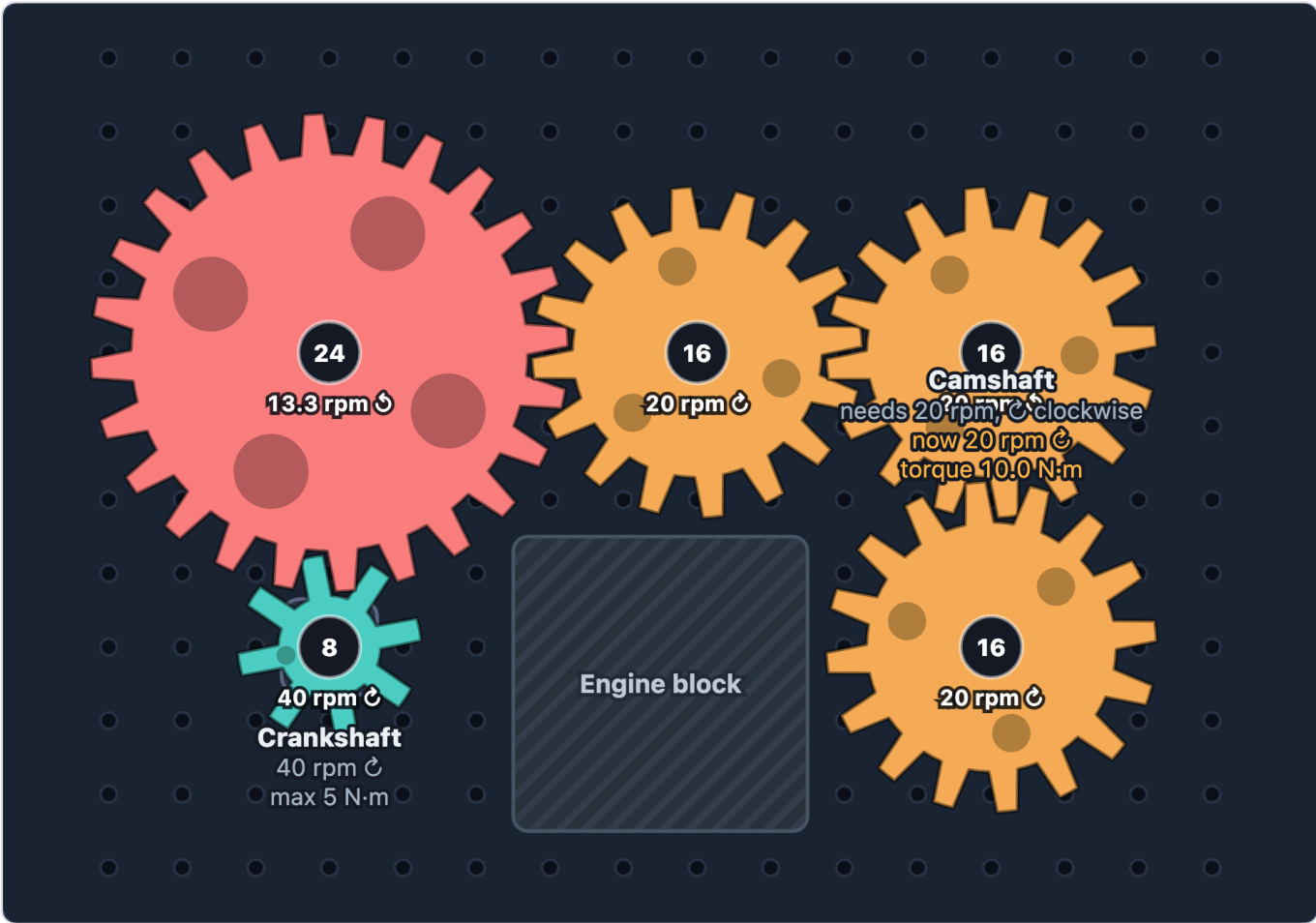
Part B • Play levels 1 to 4

Before you press **Check my gears**, predict which way and how fast the machine will turn. Record your best design for each level.

| Level | Gears I used | Cost | Stars | What went wrong, and how I fixed it |
|-------|--------------|------|-------|-------------------------------------|
| 1     |              | £    |       |                                     |
| 2     |              | £    |       |                                     |
| 3     |              | £    |       |                                     |
| 4     |              | £    |       |                                     |

Part C • Read the numbers

This is **Round the Engine**, solved, with **Show the numbers** ticked. The crankshaft's 8-tooth gear turns at 40 rpm and drives a chain of gears around the engine block to the camshaft. Each gear shows its number of teeth and its speed.



C1 The 24-tooth gear turns at 13.3 rpm. Show how this is worked out from the crankshaft's speed. [2]

**C2** The three 16-tooth gears all turn at 20 rpm. Why do they all turn at the same speed? [1]

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**C3** The camshaft turns at half the crankshaft's speed. Explain this using just the first and last gears. [2]

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**C4** There are five gears in the chain. Does the camshaft turn the same way as the crankshaft? How do you know? [2]

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### Part D • Explain

**D1** A bicycle has gears. Explain why a low gear makes it easier to ride up a steep hill, and what the rider gives up. [3]

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## Gear trains, torque and power

Name: \_\_\_\_\_

Class: \_\_\_\_\_

Date: \_\_\_\_\_

Output speed = input speed  $\times$  driver teeth  $\div$  driven teeth · torque goes up by the factor the speed goes down  
 $\omega$  (rad/s) = rpm  $\times 2\pi \div 60$  ·  $v = \omega \times r$  ·  $P = T \times \omega$  · efficiency = power out  $\div$  power in  
 In Gear Garage, an N-tooth gear has a radius of  $N \div 8$  cm on the Garage Door level.

**1 Idler gears (Round the Engine).** An 8-tooth gear on a 40 rpm crankshaft drives, in turn, gears with 24, 16, 16 and 16 teeth. The last 16-tooth gear is on the camshaft.

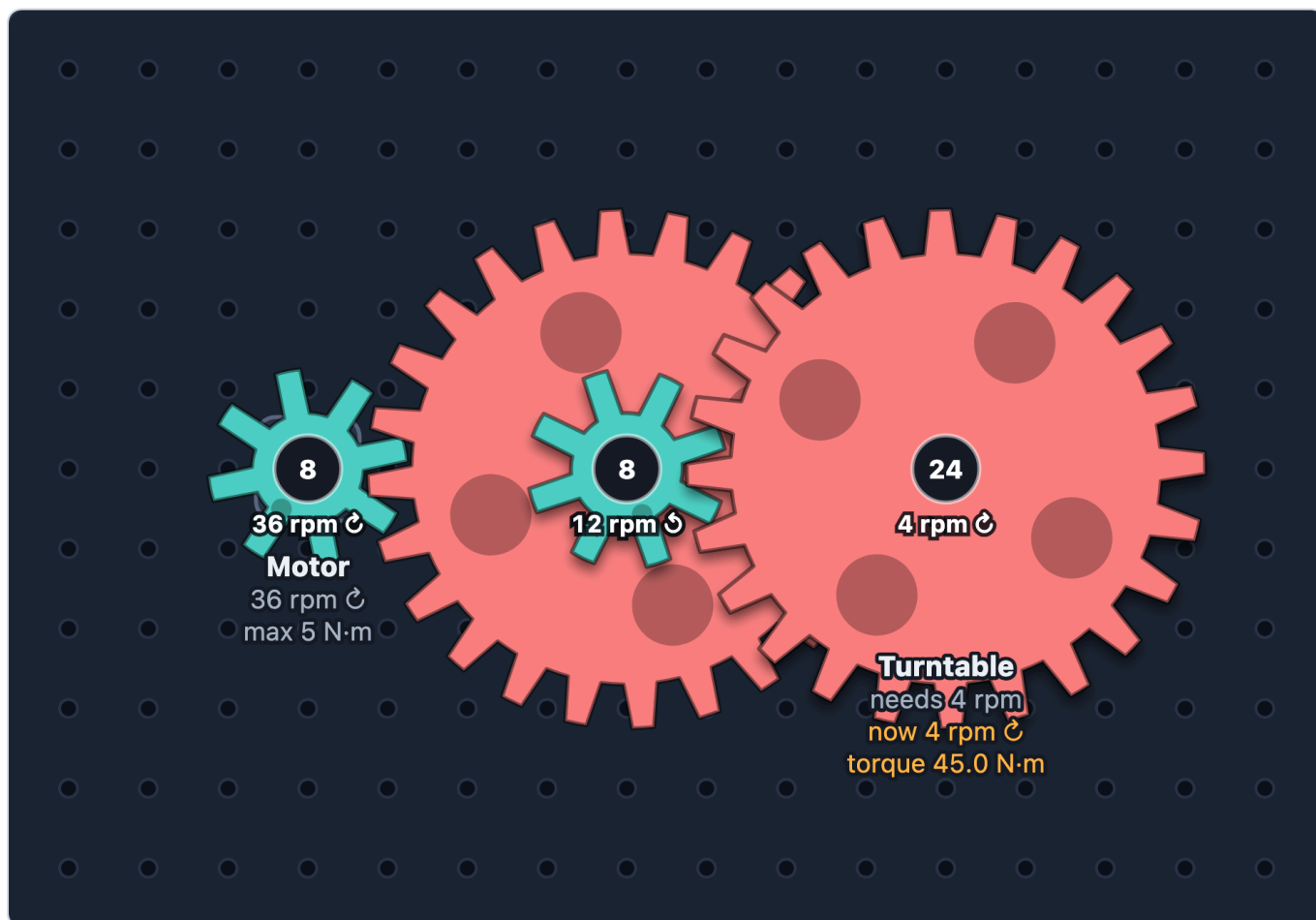
a) What is the camshaft's speed?

[1]

b) Show that the 24-tooth gear has no effect on the camshaft's speed.

[2]

**2 Compound gears.** This is the solved Compound Gears level. The 24-tooth gear in the middle has an 8-tooth gear stacked on the same axle.



a) Why does the stacked 8-tooth gear turn at 12 rpm? [1]

b) How many times slower than the motor does the turntable turn? Show that the two stages multiply. [2]

c) The motor gives 5 N·m. Ignoring friction, what torque does the turntable get? [1]

d) Using 8-, 24- and 40-tooth gears, design a compound train that turns 15 times slower. [2]

**3 Rack and pinion (Garage Door).** The motor turns at 30 rpm. The door must rise at 9–10 cm/s.

a) Convert 30 rpm to rad/s. [1]

b) A 24-tooth gear on the motor has a radius of 3 cm. How fast do its teeth move? [1]

c) Which of the 8-, 16-, 24- and 32-tooth gears on the motor gives a door speed of 9–10 cm/s? Show your working. [2]

d) In a simple chain, why doesn't the size of the gear touching the rack change the door's speed? [2]

**4 Efficiency (Every Mesh Counts).** The motor gives 1 N·m at 60 rpm. The lift needs 3.6 N·m at 15 rpm or faster. Each mesh passes on 95% of the power.

a) What is the motor's power output? [2]

b) How many times slower must the lift turn? What torque would it get with no losses? [2]

c) A compound train does this in two meshes. What torque does the lift get? [2]

d) What would a third mesh do? [2]

[6]

Continue on lined paper if you need more space.